

Eudoxa: A Trustworthy Product Recommendation Engine

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Abstract

Eudoxa is an agentic product recommendation engine using explainable, reliable reasoning and grounded facts with principled epistemics.

Agent-mediated purchase decisions are here to stay. Current agentic LLM recommendations can suffer from unreliable reasoning and hallucinated facts, reducing their trustworthiness. When higher value purchases are at stake, customers value a disciplined decision process that carefully weighs trade-offs, elicits preferences and can substantiate its recommendations in a manner that earns trust.

Eudoxa performs multi-criteria item ranking under joint fact and preference uncertainty. For this we built a knowledge-based system with a decision model that represents the uncertainty in both facts and customer preferences. The model helps a customer make trade-offs between products based on attributes. Recommendations are made using compromise ranking from multi-criteria decision analysis (MCDA). Eudoxa aims to learn more about a customer's preferences in the conversation, making the recommendation better. Customers can refine preferences in detail-oriented deliberation or be guided by generated questions/answers (prioritized by potential to change the recommendation, roughly by value of information (VOI)).

With a deep integration between a knowledge-based system and agentic LLMs, we built a more trustworthy recommendation system than one that relies solely on reasoning by an LLM. Our thesis is that LLMs and formal methods complement each other well: LLMs are flexible and naturally suited for customer interaction, while formal methods offer reliability. Over time, we expect neural training and inference to incorporate more techniques from probabilistic modeling, formal verification, and verifier-guided learning. Eudoxa uses a standard agentic LLM approach and complements it with probabilistic and decision-theoretic tools.

The decision model uses a structured algorithm as this is trusted to give a reliable recommendation. Future extensions are envisioned where a customer can make complex queries or modify the ontology with the help of an LLM, however this must be done very carefully to ensure integrity of the system.

Underlying facts are collected from multiple sources so rich context about each product is captured. All facts are collated and harmonized into a canonical schema, then calibrated through Bayesian inference using ProbLog [De Raedt et al., 2007]. The result is a database of product attributes, each annotated with a confidence and a posterior distribution. Fact provenance and original source information are maintained and are accessible to the customer via the agent.

1. Overview

1.1 Motivation

Eudoxa attempts to address three major problems in making recommendations. First, all facts are to some degree subjective and unreliable: manufacturer pages, retailers, and reviewers each have their own bias, don't cover the same ground and might fundamentally use different terminology. Second, customer preferences are vague, partial, and expressed in natural language. Third, multi-criteria trade-offs are hard: a product that is best on average may be unacceptable on some particular criterion the customer cares about.

We are open-sourcing a complete proof-of-concept recommendation workflow. The architecture extends from fact gathering and collation, to a core decision model and LLM agentic workflow with a web interface. As a demonstration of our approach, we included an example for making snowboard product recommendations.

1.2 Beyond LLM-only recommendations

An LLM-only agentic recommender system addresses all three problems with the same instrument: a language model reasoning directly over retrieved text. This is flexible but structurally uncalibrated: the model has no explicit representation of *how much* it believes each fact, no replicable reasoning process, and limited artifacts a skeptical customer can audit (CoT will reveal only high-level rationalizations). Furthermore, LLMs' reasoning process is not well understood and is known to be prone to hallucinations. This problem becomes more pronounced when covering larger datasets where a LLM's context window can't cope with the data volume.

Many modern LLMs are optimized for agentic workflows and tool use. We have observed that agentic solvers in complex domains do the best when discovering and using pre-existing tools suitable to the problem at hand (e.g. copying or regurgitating known solutions, or using known code libraries). A related approach is MCP, where LLMs are presented with APIs to known working code and instructions on how to use the API.

Eudoxa's recommendations are based on an explicit decision model exposed to the LLM through an agentic harness, which fits the tool-use pattern at which current models excel. Considerable research is directed at improving LLM reasoning itself, including reinforcement learning from verifiable rewards (RLVR) against formal tools such as the Lean proof assistant, and mechanistic interpretability studies have found that trained models can develop internal circuits implementing recognizable algorithms. We expect LLM reasoning to continue improving in both capability and reliability, but a substantial gap remains between implicit, learned reasoning and explicit, verifiable computation. Eudoxa's architecture is therefore a sound compromise for building trustworthy systems today.

1.3 Eudoxa's architecture



1.3.1 Decision Model overview

Consider a large, established product catalog with many attributes, surrounded by a body of external context: corroborating, conflicting, and complementary information, reviews, and the like. For an LLM, processing and reasoning reliably over a large input is hard, especially when the customer demands accurate attention to very specific criteria. LLM performance degrades over large contexts, and in our testing of ad-hoc, one-shot LLM recommendations we observed many failure modes.

Eudoxa's decision model represents both customer preferences and product attributes as random variables, with each distribution derived from the available evidence: calibrated facts on the attribute side, elicited statements on the preference side. Recommendations are computed by a joint Monte Carlo simulation over these distributions: each sampled possible world is ranked with VIKOR compromise ranking, and results are aggregated across worlds following Stochastic Multicriteria Acceptability Analysis (SMAA-2 [Lahdelma and Salminen, 2001]). The output is therefore a ranking together with a quantified statement of its robustness.

The decision model is designed to help a customer make trade-offs between attributes of products. The LLM handles natural-language interaction and encodes preferences expressed by the customer as parameters to the decision model. Product recommendations are strictly handled by a deterministic, verifiable algorithm. The LLM has direct access to facts in the ontology, and can answer ad-hoc queries about them. Complex queries that go beyond preference-based recommendations from the decision model fall back to the LLM's reasoning. Guardrails with explicit instructions limit what the LLM may say in that situation; how restrictive they need to be depends on the particular use case for Eudoxa. In future we expect more comprehensive tooling into the ontology to be exposed to the LLM.

1.3.2. Ontology overview

In production, a recommendation system benefits greatly from pre-processing its data: the cost is paid once and amortized over many customer queries. The value of that pre-processing depends on how far it distills: the more the raw material is condensed into a compact, information-dense form, the cheaper and more reliable every subsequent query becomes. An ontology is particularly efficient: by query time, every fact has already been normalized, reconciled across sources, and annotated with confidence and provenance. An LLM-only system could do some of the same work upfront, but its query-time interpretation and reasoning would still run over a large textual context, with the failure

modes described above, while a conventional algorithm operating on the pre-processed data is faster, cheaper, and replicable.

The failure modes of an ontology- and model-based system should not be underestimated. The price of its scalability, speed, and reliability is inflexibility and brittleness: errors and modeling assumptions baked in during construction are very hard to correct at query time. The system also cannot dynamically incorporate new information; regularly re-running the ontology construction pipeline mitigates this, but a re-run is slow and cannot happen while serving a customer conversation. The very policy that grounds the system, trusting only the calibrated facts in the ontology, suppresses dynamic reasoning at query time. Eudoxa's design accepts these trade-offs in exchange for reliability and trustworthiness. Quality assurance during ontology construction is therefore important, and fortunately implementers have much more control here than over the reasoning abilities of an LLM.

Eudoxa's current implementation uses LLMs in ontology construction, but largely as a convenience for the proof of concept implementation. For a production system these tasks should be assigned to domain experts, or at least run under expert supervision. Where the LLM is used, the work is divided into small, well-bounded tasks, each with confined context and complexity, so that hallucination is both less likely and limited in impact. LLMs are used for linguistic tasks such as reading sources, normalizing vocabulary, and translating sourced claims into formal facts; these tasks require general knowledge rather than trusted judgment. Epistemic tasks are never left to the LLM. Reconciling contradictory claims, weighing agreement and disagreement between sources, assigning degrees of confidence to sourced information, tracing provenance, and propagating uncertainty are all encoded in probabilistic logic and executed deterministically in ProbLog.

1.3.3. Information Broker and LLM agent overview

Eudoxa's UX demonstrates a complete proof-of-concept recommendation workflow. It is built on an agent-and-broker architecture: a conversational LLM handles the dialogue, while a privileged broker has access to the raw data, the catalog, and the decision model, exposing them only as typed, validated tools to the LLM. The separation is a firewall: the LLM can invoke the decision model but cannot encroach on it, bypass it, or second-guess its output, and it receives only what the broker exposes. How strict this separation needs to be remains an open question: the implementation supports transparency modes ranging from black-box (internal diagnostics redacted, so the LLM can neither leak nor confabulate based on them) to fully transparent. The usefulness of this strict approach is still under qualitative evaluation.

(The released code also includes visualization tools for developers that illustrate the decision model's workings directly).

2. Ontology construction

In Eudoxa the ontology is a pipeline of data extraction and aggregation of product facts leading to a database of product attributes that are relevant to a recommendation decision. We have included an example ontology useful for making snowboard recommendations for an online merchant. The released source code can be extended to offer multiple ontologies and product domains.

An ontology is constructed up-front, this amortizes the construction cost over serving the decision model's recommendations. We envisage regular batch processing to update the ontology, and limited LLM-assisted modifications, for example incorporating epistemic feedback from the customer about their stated trust or distrust in particular sources.



2.1 Sourcing product attributes

The ontology is seeded with a given product category or given list of products. Per product facts are then collected by an LLM with web-search tooling, operating under a system prompt that enforces a structured output contract. Every attribute claim is emitted with its citations, and each citation carries the model's assessment of three things: how directly the cited passage supports the claim, how credible the source is for a claim of that kind (a manufacturer's specification, a retailer's copy, and a forum post are treated differently), and how much the attribute matters to a purchaser. The prompt instructs the model to preserve disagreement rather than resolve it, e.g. if two sources give different flex ratings for the same snowboard, both claims are emitted and flagged as conflicting. The conflicting claims will be propagated through Bayesian inference and result in lower confidence and higher standard deviation for the sampling interval in the recommendation algorithm for the underlying attribute. (Future work could model this as a categorical random variable). Difficulty in parsing or translating attribute values from the source text will result in an attempt to attach a value, which will be tagged as a very low confidence guess (which is handled appropriately by the inference and later the recommendation engine).

In this demonstration the sourcing is automated, with manual spot checks. In various ways the LLM's function and context are tightly constrained for task-specific actions. This substantially reduces extraction errors. To further reduce the probability of errors, a human expert supervision step can be added. Because Eudoxa natively handles uncertainty of facts, imperfect extraction can be well mitigated. Nevertheless, any error that survives this stage is persistent: it will be served, unchanged, until the pipeline is re-run. While mistakes are durable, they are also localized, detectable, and correctable. This stands in contrast to an error in an LLM's embeddings or reasoning, which is more transient and unlikely to be identifiable or correctable.

2.2 Harmonization

Different sources name the same property differently (e.g. a snowboard's "flex", "flex rating", "stiffness 1–10"). A harmonization pass uses an LLM to cluster variant attribute names across the whole corpus into generic canonical attributes. The mapping is also joined back to the raw facts. The output is a canonical attribute table in which every raw claim is tagged with its canonical attribute but retains its original value, source, and citation assessments; nothing is discarded or averaged at this stage so later inquiry by the customer or verification algorithm has full visibility.

The LLM's role in harmonization is vocabulary unification. The whole unification uses a map-reduce style algorithm to keep batches small for manageable LLM context. (A later join on data itself uses regular code). As in the sourcing step, a human domain expert supervision step can be added to ensure quality.

2.3 Calibration and verification via probabilistic logic

The calibration stage is the epistemic core of the system. For each product, the harmonized claims are compiled into a probabilistic logic program: an explicit epistemic model, written by an LLM in the form of a ProbLog program. The program encodes which candidate values are in play, how reliable each source tends to be for each kind of claim, which sources are independent and which merely echo one another (retailers repeating a manufacturer's spec sheet), and how the observed claims constrain one another.

Example ProbLog extract where there is source disagreement about a snowboard's flex rating (how soft a board is). More example details are available in the released code.

```
0.82::accurate(s_merch, flex_rating_10). % source merchant: is likely accurate with probability 0.82.
0.88::accurate(s29, flex_rating_10). % source 29: the manufacturer is likely accurate with probability 0.88.
0.78::accurate(s26, flex_rating_10). % source 26: review site is likely accurate with probability 0.78.

% @attr flex_rating_10 @type numeric @values v3_5=3.5 v6=6.0 v6_5=6.5
% Loose estimated prior distribution – updated to posterior via conditioning on evidence(all_consistent(...)).
0.30::true_val(flex_rating_10, v3_5); 0.40::true_val(flex_rating_10, v6); 0.30::true_val(flex_rating_10, v6_5).

measured(s_merch, flex_rating_10, v3_5). measured(s29, flex_rating_10, v6). measured(s26, flex_rating_10, v6_5).
all_consistent(flex_rating_10) :- consistent(s_merch, flex_rating_10), consistent(s29, flex_rating_10), consistent(s26,
flex_rating_10).
evidence(all_consistent(flex_rating_10)). % Conditions the model

% print out posteriors
query(true_val(flex_rating_10, v3_5)). query(true_val(flex_rating_10, v6)). query(true_val(flex_rating_10, v6_5)).
```

Probabilistic inference over this model produces, for every attribute of every product, a value, confidence in the value and a posterior distribution over values. When sources agree, the posterior is peaked and confidence is high. When they disagree, the disagreement is not collapsed by majority vote or by an LLM's judgment call; it survives as a spread-out posterior, and that spread propagates into the decision model as value uncertainty. (A future improvement would be to model this as a categorical random variable with a set of finite values).

Flex_rating_10 example from above, calibrated:

```
{
  "name": "flex_rating_10",
  "type": "numeric",
  "posterior": {
    "6.5": 0.214,
    "6.0": 0.524,
    "3.5": 0.262
  },
}
```

Which is mapped the following attribute (p.s. other mapping approaches are also available in the code):
flex_rating_mean = 5.45, flex_rating_std = 1.18, flex_rating_conf = 0.524

The LLM is confined to authoring the epistemic model: priors, reliability estimates, and evidence structure as a ProbLog program. Any LLM error is confined to an artifact that can be read and audited. Inference over that model is deterministic probabilistic reasoning and every posterior is thus reproducible.

2.4 Aggregation into the comparison database

A shortlist of product attributes are picked that are most relevant to a purchase decision. This can be done by a human expert or a supervised prompted LLM. (Where sources express a property as text rather than numbers, a text-to-number mapping is produced as the decision model works only with numbers so it can sample under uncertainty. Values with no mapping are left blank rather than guessed, and coverage gaps are modeled explicitly by using the mean and an extended(*) standard deviation over all data for the attribute. (* to account for altogether unseen outliers).

An excerpt from the snowboard example database shows scalars with a mean, std, confidence. The sidecut_type and shape are categorical values.

product_guid	0wlvYbEX2	2LdthH7GBg	Cge1q0e1p0	Fw044ugEFM	aiSeM4Zayv
title	Mountain_Twin_Pro	Freecarver_9000s	Assassin	Skate_Banana	Freecarver_6000s
rider_level	Advanced_Expert	Advanced	Intermediate_to_Advanced	Beginner_to_Pro	Intermediate_to_Expert
brand	Jones_Snowboards	Jones_Snowboards	Salomon	Lib_Tech	Jones_Snowboards
shape	Directional_Twin_3D_Contour	Tapered_Directional	Directional_Twin	Twin	Tapered_Directional
model_year	2027	2027	2026	2027	2027
price_mean	1199.99	1149.99	719.99	969.99	1149.99
price_std	115	176	115	307	115
price_confidence	0.95	0.88	0.99	0.63	1
sidecut_type	Progressive_Sidecut	Progressive_sidecut_radius	Quadralizer_Sidecut		Progressive_Sidecut
flex_rating_mean	8	8	5	4	6
flex_rating_std	0.66	1.25	0.87	1.2	0.45
flex_rating_conf	0.89	0.6	0.81	0.63	0.99
sidecut_radius_159_mean	8	9	8	8	7
sidecut_radius_159_std	0.35	0.44	0.35	0.35	0.35
sidecut_radius_159_conf	0.99	0.92	0.99	1	1
effective_edge_mean	121	130	116	119	128
effective_edge_std	2	2	2	47	2
effective_edge_conf	0.99	0.92	0.99	0.2	1
waist_width_mean	25	27	25	26	26
waist_width_std	0	0	0	0	0
waist_width_conf	0.92	0.92	0.99	1	1

3. The decision engine & recommendations

Customer preferences are encoded as an explicit spec and the recommendation is computed by a single joint Monte Carlo simulation over preference uncertainty and data uncertainty. (See the example visualization in the appendix). The decision model makes a product recommendation in the form of a ranking, answering two questions at once: what is the best compromise given what we currently believe, and how much confidence does it have in that recommendation.

A human expert or prompted LLM can optionally make an initial rough guess at customer preferences with high uncertainty. These may be based on expected sentiment for average customers, or might be based on prior knowledge for a particular customer. The model can be run with no preferences (see '*dormant*' below).

3.1 Encoding preferences

Customer preferences are captured in a criteria specification: a typed schema that defines, for each criterion, what kind of value it takes, which target value (or value direction) is preferred, and how important it is to the customer. The latter is encoded as an attribute weight which is used by the recommendation engine to scale the score of each product on that attribute.

When a customer expresses preferences (or clicks on a ready-made picker in the UI), the LLM will encode this as a preference change and compile it into a small algebra of validated transforms: adjust a weight, flip a direction, set a target, activate or dismiss a criterion. When a customer says "I don't really care about length, but it must handle ice," (*) the LLM's job ends at emitting transform calls. (*) in this case, `length` becomes an *excluded* attribute, and `ice_performance` gets a high weighting).

Application of the preference spec is deterministic, and malformed or out-of-schema transforms are rejected by validation. This is an additional safeguard to constrain and validate LLM output.

Each criterion also carries an explicit lifecycle state:

- **Active:** the customer has expressed interest; the criterion participates in the ranking.
- **Dormant:** the customer has said nothing about it; the system treats it as an open question.
- **Excluded:** the customer has explicitly dismissed it; it is dropped from the analysis entirely.

Every criterion in Eudoxa is either active, dormant, or excluded. Active criteria drive the ranking and are refined through targeted questions; excluded criteria are dropped entirely. Dormant criteria are treated as genuinely unknown: the engine samples plausible weights and target values for them, and a discovery algorithm surfaces the ones whose activation could change the outcome. Separating "not yet expressed" from "rejected" is what makes discovery possible and guides preference elicitation from the customer.

3.2 Sampling possible worlds

Each iteration of the simulation draws one complete "possible world." Preferences are perturbed (tightly when firm, diffusely when provisional), and dormant criteria receive a bounded share of influence, representing unexpressed preferences without letting them dominate. When sampling facts, each value is drawn from its calibrated distribution, so a value on which sources agree barely moves, while a contested or low confidence one swings across its plausible range.

The injected noise is not a modeling convenience: it is the uncertainty that the calibration stage measured from source disagreement, so the simulation is empirically grounded end to end.

Each Monte Carlo iteration draws values for weight preferences and attribute values:

Weights are drawn from a Dirichlet distribution centered on the user's current weights, with a concentration set by a perturbation parameter — tight when preferences are firm, diffuse when they are provisional. Dormant criteria are handled by hybrid sampling: a sub-budget drawn from `Uniform(0, dormant_budget)` is allocated across the dormant criteria by a uniform Dirichlet, so that

unexpressed preferences enter some worlds at realistic magnitudes without contaminating the main ranking (the deterministic ranking is computed with `dormant_budget` set to zero).

Values are drawn from distributions parameterized by the pipeline's confidence estimates. Scored and categorical values use mean-preserving Beta distributions whose spread narrows with confidence: a fully agreed-upon fact barely moves, while a contested one swings. Numeric values use lognormal or normal distributions as appropriate.

Common random numbers are used in the discovery phase to compare counterfactual scenarios. Three values (min, mean, max of the range) are drawn for each unexplored criterion and evaluated for ability to disrupt rankings.

3.3 Compromise ranking

Within each sampled world, products are ranked with VIKOR compromise ranking [Opricovic and Tzeng, 2004]. This judges each product on both its overall weighted performance and its single worst shortfall, and balances the two: a product that is excellent on average but severely deficient on a criterion the customer cares about cannot win by compensation. (A model parameter allows one to trade off "best on balance" vs "fewest dealbreakers.")

3.4 Acceptability across worlds

Across all sampled worlds, Eudoxa accumulates a ranking profile ("rank-acceptability" in the manner of SMAA-2 [Lahdelma and Salminen, 2001]): for each product, how often it finishes first, second, third, and so on. This profile shows directly whether the leader wins in 85% of plausible worlds or merely 30%. For easy ordering, the Holistic Acceptability Index (HAI) collapses rankings into a single score by giving each finishing position a weight and summing with a decaying weight [Barron and Barrett, 1996]. Products whose indices differ by less than the simulation's statistical noise are treated as tied; when the top spot cannot be separated by secondary criteria, the system reports the tie and lists the interchangeable winners.

3.5 Recommendation confidence

The confidence attached to a recommendation is a statistic of the simulation. It reflects how consistently the winner prevails across possible worlds (Monte Carlo iterations), how clearly it separates from the runner-up, and how much of the preference space remains unexplored.

Because trust in a source is an explicit quantity, the architecture can be extended to allow epistemic control. A future version could let a customer adjust the credence given to manufacturers, reviewers, or other sources and observe how the recommendation responds. This allows further personalization of the recommender to a customer and could be added to the preference encoding.

4. Interaction: questions & exploration

4.1 Model generated follow-up questions

As part of running the decision model, Eudoxa searches its own simulation for the questions worth asking: those whose answers would materially move the recommendation. **Refinement questions** are suggested by the decision model to the LLM agent, and aim to refine what the customer has already expressed: preferences whose influence on the outcome is unstable, facts whose uncertainty is decision-relevant, pairs of products whose ordering flips from world to world, and criteria on which the current leader is fragile. **Discovery questions** probe what the customer has not yet said: if giving weight to a dormant criterion frequently lets an outsider break into the top ranks, or reshuffle the current leaders, the customer is asked whether they care about it. Each candidate question carries an estimate of how much its answer would shift the recommendation (**impact_score**, roughly = value of information). Only the highest-impact questions are handed to the conversational LLM for phrasing.

The LLM-agent will prompt the customer with these questions verbatim, or encode in an easy-to-click “fast response” button. Any answers (or selections) by the customer must be translated by the LLM-agent into customer preferences using the preference spec and these are fed into a new simulation run.

4.2 Recommendation exploration

The LLM-agent exposes recommendation results and the source information and database entries of the ontology to the LLM via tool definitions. (The level of detail in the decision model output is subject to a firewall, described below).

Examples of data tool definitions are `get_product_grounding_evidence`, `get_product_source_text`, `get_product_full_attributes`, `search_products`, `list_products`.

Using these tools, the customer can drill down from any recommendation through every layer of its evidence: to the alternative recommendations and their acceptability profiles, to the per-criterion contributions behind a product's score, and to the calibrated value and confidence of any attribute. The original quotes from the sources with URL, retrieval date, credibility score, and written justification are also available.

Each recommendation run is repeatable: a random seed is used for sampling for all random variables, thus the Monte Carlo results will be the same given the same set of input preferences.

Customer-driven explorations over possible preference trade-offs are answered by re-running the decision model rather than by LLM speculation.

5. Brief overview of Eudoxa serving architecture

Eudoxa has an agent and broker component. The agent runs the conversation with the LLM and exposes tools to the LLM. These tool calls are relayed to the broker which executes them as a privileged operation (e.g. loading data, validating preferences, running the decision model, querying the cache) and returns a (sometimes redacted) result to the agent.

In the evaluation of Eudoxa, we explored ways of putting guardrails around the LLM. One mechanism is a firewall between the broker and the LLM. This aims to keep as much information about the recommendation algorithm and its internal workings as possible from the LLM. The LLM system prompt merely describes it as a ranking and recommendation engine and it's given a schema for encoding customer preferences.

The firewall is to test the hypothesis that an LLM with access to detailed internal results and internal algorithms will start reasoning over them: recomputing scores, second-guessing the ranking against its own intuitions, or guessing at the algorithm and explaining the result in terms of its guess. To evaluate the value of this firewall, two transparency modes are supported: in *black-box* mode the broker redacts internal diagnostics from what the LLM sees, and in *transparent* mode the full decision structure is available to the LLM. (A *debug* mode exposes even more internals to the LLM). We found in practice that it was slightly easier for the customer to reason with the LLM-agent over specific technical aspects of a recommendation when run in “transparent” mode, but it was a minor difference. Anecdotally, we did not find a degradation in the recommendation performance of the model when run in black box mode, though we have not yet run a rigorous evaluation. When pushed, the LLM will not reveal or second-guess the algorithm behind the recommendation engine, because it has been instructed to do so, and because it doesn't have full access to the model internals. We have not done ablation to test the effectiveness of each.

6. Eudoxa versus an LLM-only agentic recommender

An LLM-only approach is stronger for:

Schema-free reasoning: an LLM can weigh a consideration no one encoded, while Eudoxa relies on a fixed ontology (though it could be re-built regularly) and a fixed recommendation algorithm.

Flexible constraints and interactions: Eudoxa's additive scoring treats requirements as weights, so complex interactions between features are not modeled unless explicitly anticipated during ontology construction and encoded; an LLM handles both natively.

Eudoxa is stronger for:

Principled elicitation: refinement and discovery questions are ranked by simulated impact on the decision and with a principled statistical basis, so have sound justification.

Scale: large product databases plus extensive attributes can easily swamp the effective operating size of an LLM's context window. A database and a deterministic, scalable algorithm handle it much better.

Veto sensitivity: the regret-sensitive ranking structurally prevents the averaging failure LLMs are prone to when a product is excellent on most axes.

Calibration: source disagreement is measured and propagated as probability rather than adjudicated implicitly by a generative model.

Auditability: every recommendation decomposes into inspectable artifacts, so "why this product?" can be reliably answered.

Amortized cost: sourcing, calibration, and aggregation run once, and their cost is amortized over all subsequent recommendations. An LLM agentic recommender pays for retrieval and reasoning on every query, and it uses much more expensive computing resources.

A caution on evaluation. We found Eudoxa fared well in manual head-to-head evaluations when compared to LLMs, when evaluated by domain experts and/or very attentive users. When attempting larger scale evaluation using LLM-as-judge, we found more than a few instances where the judge had failure modes similar to the LLM being evaluated. For example, the LLM-as-judge would not rigorously consider all customer preference when making a recommendation. Similarly, where there was nuanced or complex source disagreement about a fact, the LLM-as-judge would disagree with Eudoxa because it did not do the careful calibration of a fact's provenance. This was particularly noticeable when a large dataset was used. This meant a mechanical evaluation at large scale was not very reliable. We're confident that it's possible to do better, and hope to improve this in future.

7. Conclusion

Eudoxa is a deep integration between a knowledge-based system and agentic LLMs, building a more trustworthy recommendation system than one that relies solely on reasoning by an LLM.

It is built on the thesis that LLMs and formal methods complement each other: the LLM supplies the flexibility that natural customer interaction demands, while grounded facts and explicit probabilistic reasoning offer the reliability that earns trust when higher-value purchases are at stake. This remains the advantage of traditional formal methods while current LLMs still regularly exhibit failure modes.

Where the flexibility and general knowledge of LLMs are used to make ontology construction easier, it is in relatively small, well-bounded linguistic tasks. Quality assurance during ontology construction is a place where implementers have real leverage and control over the input into a deterministic recommender. This is a higher degree of control than is possible over a pure LLM recommender.

Neural and transformer-based models are expected to continue improving, along two paths relevant here: training techniques in which formal methods play a central role, and richer ensembles of AI models that incorporate, or entirely subsume, components with a traditional probabilistic heritage. Bengio's Scientist AI, a non-agentic predictor trained to approximate a Bayesian posterior over natural-language statements, is an early example of the latter [Bengio et al., 2026].

While there is still a clear split between the decision model, which produces explicit product rankings, and a fallback to LLM reasoning for ad-hoc queries over facts, there is potential in extending the blend of formal methods with LLM guardrails to flexible queries. Engineering investment can make the ontology construction pipeline responsive enough to dynamically incorporate epistemic feedback from the customer, for example a customer's stated trust or distrust in particular sources.

Product recommendation was chosen as Eudoxa's focus because it is a timely use case and one that benefits greatly from trustworthiness. The approach, however, applies to a wide range of decision problems in which preference trade-offs and rankings of alternatives are central to the user experience.

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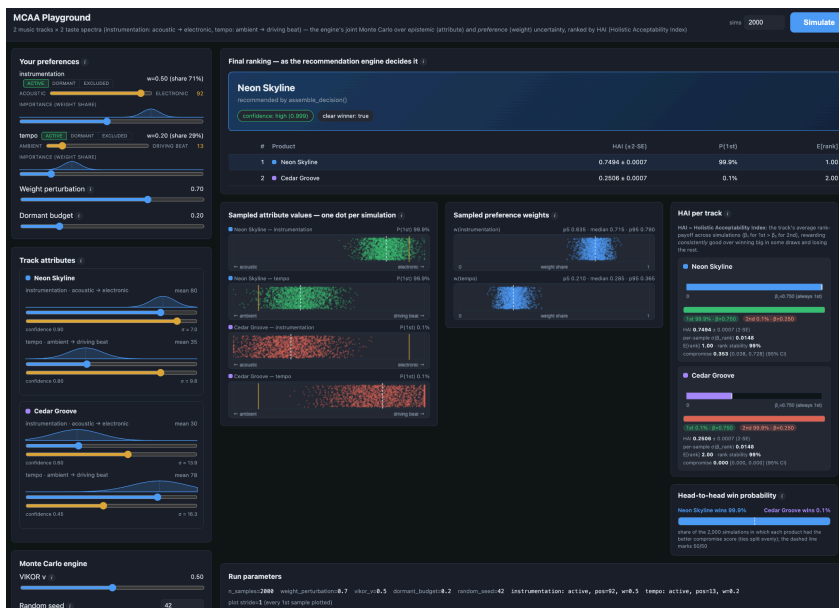
Appendix

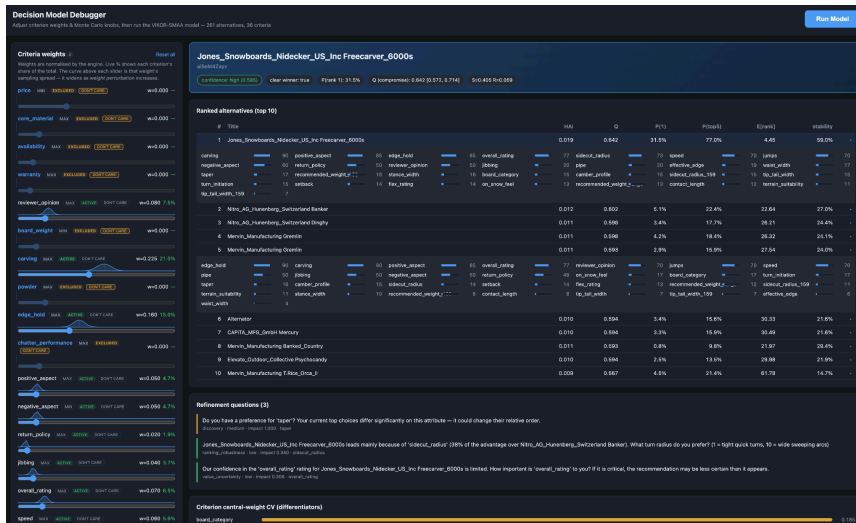
A screenshot of Eudoxa during a recommendation flow:

The screenshot shows the Eudoxa interface with a user query: "I'm interested in a cruiisy carving boards, price is not a problem, it must have great edge hold. I don't care about going in the park or jibbing." The interface displays a detailed recommendation for the GNU Banked Country board, highlighting its 39% probability of being the top pick and its excellent carving and edge hold. It also provides a comparison table for three boards: GNU Banked Country (2026), GNU Gremlin (2026), and Lib Tech Dynamo. The table includes columns for board name, price, score, P(#1), ranking driver, relative strength, and regret driver. The GNU Banked Country is the clear winner with a 60% confidence level. The interface also includes a sidebar for session details, latest decision, and preference overrides, and a bottom section for user input and a send button.

#	BOARD	PRICE	SCORE	P(#1)	RANKING DRIVER	RELATIVE STRENGTH	REGRET DRIVER
#1	GNU Banked Country (2026)	AUD \$1,049.99	77%	39%	carving, edge hold	reviewer opinion, carving (32%)	carving (32%)
#2	GNU Gremlin (2026)	AUD \$969.99	74%	14%	carving, edge hold	positive aspect, powder	chatter performance (54%)
#3	Lib Tech Dynamo	AUD \$969.99	74%	3%	carving, edge hold	reviewer opinion, chatter performance	edge hold (48%)

A screenshot of a developer visualisation tool showing a full Monte Carlo simulation: 2000 samples of the random variables representing preferences and product attributes, together with the aggregated ranking that results:





Comparison to ChatGPT

This is by no means a thorough comparison, but illustrative of how details like disagreement about a product attribute are easily overlooked when an LLM is trying to aggregate lots of data on the fly, and not explicitly forced to reason about all facts. (ChatGPT 5.5)

User prompt: You're a snowboard expert. I'd like to buy a snowboard for all-mountain riding. It should have good edge hold, and do equally well for jibbing and carving. Powder riding is very important. Also make sure they are well reviewed boards. Very important is a soft flex board. Please include the Burton Process to the comparison, I like the look of it. Give me a ranking table for the 6 boards with attributes including flex (also powder, carving, jibbing, edge hold, reviewer opinion)

Rank	Board	Flex	Powder	Carving	Jibbing	Edge Hold	Reviewer Opinion
1	Salomon HPS Louif Paradis	5.5-6/10	★★★★★	★★★★★	★★★★★	★★★★★	9.7/10
2	Burton Process (Camber)	3.5-4/10	★★★★★	★★★★★	★★★★★	★★★★★	9.3/10
3	CAPiTA Outerspace Living	4/10	★★★★★	★★★★★	★★★★★	★★★★★	9.2/10
4	K2 Almanac	4-5/10	★★★★★	★★★★★	★★★★★	★★★★★	9.0/10
5	Bataleon Whatever	4-4.5/10	★★★★★	★★★★★	★★★★★	★★★★★	8.9/10
6	CAPiTA Spring Break Slush Slasher	3/10	★★★★★	★★★★★	★★★★★	★★★★★	8.8/10*

User prompt: Remember that I actually asked for a soft or medium soft flex?

ChatGPT forgot the soft flex criteria.....

Yes—and that's an important correction. Looking back at your original brief:

"Very important is a soft flex board."

I let the HPS Louif Paradis climb to the top because of its powder performance and rave reviews, but if I strictly honour **all** of your requirements, especially **soft or medium-soft flex**, then it shouldn't be #1 anymore.

Rank	Board	Flex	Powder	Carving	Jibbing	Edge Hold	Review Consensus	Fit for You
1	K2 Almanac	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	9.6 / 10
2	Burton Process Camber	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	9.4 / 10
3	CAPiTA Outerspace Living	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	9.2 / 10
4	Salomon HPS Louif Paradis	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	9.1 / 10
5	Bataleon Whatever	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	8.9 / 10
6	CAPiTA Spring Break Slush Slasher	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	8.4 / 10

As one can see, the Burton board gets 5 stars because it's nominally soft flex, but see Eudoxa source disagreement for comparison - there's disagreement over the flex.